$$\Delta\boldsymbol{\phi} \in \mathbb{R}^4, \quad \Delta\mathbf{q} \in \mathbb{R}^{12}$$

$$a_t = [\Delta\boldsymbol{\phi}, \Delta\mathbf{q}] \in \mathbb{R}^{16}$$

$$\psi_l = \phi_l + \Delta\phi_l + \Delta\phi_0$$

$$q_{i\in l}^{\text{target}} = \text{IK}[p(\psi_l)] + \Delta q_{i\in l}$$